Golden Dataset & Analytical Pipeline

Collections Recovery Performance Investigation

Purpose: establish a trustworthy analytical layer so recovery performance can be measured independently of duplicated, inconsistent or misleading source data.

1. Objective

The Golden Dataset creates a reproducible analytical foundation from multiple collections systems. It standardizes source data, identifies data-quality issues, applies controlled deduplication and attribution rules, and produces validated recovery data for Python, Power BI and executive reporting.

2. Analytical Flow

RAW DATA → STAGING → CLEAN → DATA FORENSICS → GOLDEN → FEATURES / METRICS → PYTHON + POWER BI

3. SQL Server Analytical Processing

SQL Server is the core transformation, validation and metric-calculation layer in the solution. The operational CSV files were imported into SQL Server and processed through reproducible SQL steps before the validated analytical outputs were consumed by Python and Power BI.

SQL workflow: source tables → staging standardization → duplicate and integrity checks → clean analytical tables → Golden recovery dataset → recovery/DPD/risk metrics → Python and Power BI.

SQL activity	Purpose in the assignment
Import and staging validation	Confirm source tables and row counts; standardize data types and formats.
Primary-key duplicate checks	Identify duplicate IDs across the 17 operational datasets without deleting records prematurely.
Record-level forensic inspection	Distinguish exact duplicates from legitimate history, retries and identity collisions.
Referential and attribution checks	Validate account, borrower, call, attempt and payment relationships.
Clean-layer construction	Create standardized analytical tables for consistent downstream reporting.
Payment validation	Deduplicate payment records and isolate validated SUCCESS transactions.
Golden recovery construction	Create the trusted recovery population used for the recovery KPI.
Analytical metrics	Calculate recovery, recovery rate, DPD/risk analysis and monthly MoM performance.
Claim validation	Independently test the reported 11% month-on-month recovery improvement.

Key SQL outputs: 25,500 raw payments → 25,000 clean payments; 60,600 WhatsApp events → 60,000 clean events; 17,534 validated recovery transactions; ₹1.316B validated recovery.

Raw operational data is first standardized in staging. The clean layer applies type normalization, duplicate handling, missing-value checks and integrity validation. Forensic checks are then used to identify attribution and identity risks before the Golden recovery population is created.

4. Source-of-Truth Decisions

Area	Source of truth	Reason
Accounts	clean.accounts	Account-level source for outstanding amount, principal, DPD, risk, loan type and status.
Payments	clean.payments	Used for payment status, payment method and payment-quality analysis.
Recovery	gold.recovery_payments	Authoritative validated successful-payment population used for recovery reporting.
Payment ID	payment_id	Operational payment identifier after duplicate-copy handling.
Borrower attribution	account_id → accounts.account_id → accounts.borrower_id	Preferred because 25,499 borrower/account mismatches were identified.
Call events	attempt_id preferred; call_id retained as reference	call_id showed cross-system inconsistencies and cannot safely be assumed to be a unique event key.

5. Cleaning Impact & Forensic Findings

Dataset / issue	Observed result	Treatment	Business implication
Payments	25,500 raw → 25,000 clean; 500 rows removed	Remove duplicate copies	Reduces risk of overstated recovery.
Payment references	382 / 25,000 missing (1.53%)	Retain NULL; no imputation	Reference-based reconciliation is incomplete for these records.
Borrowers	8,566 duplicate borrower IDs; 19,585 excess records	Do not auto-delete	Identity/history requires further resolution.
Agents	1,000 duplicate agent IDs; 29,000 excess rows	Do not auto-delete	Person-level agent attribution is unreliable.
Agent impact	88,241 calls; 28,305 accounts; 15,000 sessions affected	Flag attribution risk	Agent performance comparisons require identity remediation.
Calls / attempts	call_id inconsistencies across accounts, borrowers, times, agents and vendors	Prefer validated attempt-level key	Call-level event attribution can be misleading.
WhatsApp	60,600 source → 60,000 clean; 600 exact duplicates removed	Remove exact duplicate copies	Prevents digital-event volume inflation; 0 orphan events remained.

6. Validated Recovery Dataset

Validated recovery	Recovery transactions	Recovered accounts	Recovered borrowers
₹1,315,583,965.60	**17,534**	**13,284**	**7,987**

Validated Recovery Rate = Validated Successful Recovery ÷ Outstanding Exposure. Only validated successful payments are treated as recovery; failed, pending, reversed and duplicate events are not treated as recovered cash.

7. Golden Dataset Business Purpose

- Provides a controlled and reproducible source for recovery reporting.
- Reduces the risk of duplicate or inflated recovery metrics.
- Makes payment attribution explicit and traceable.
- Separates observed facts from unresolved identity or causal limitations.
- Feeds the Python notebook and Power BI dashboard from the same analytical foundation.

8. Conclusion

The Golden Dataset establishes a consistent analytical source of truth for collections recovery. It preserves legitimate transactional history while correcting confirmed duplicate-event problems and explicitly documenting unresolved borrower, agent and call-identity risks. This creates the foundation required to independently test reported recovery performance.

Key principle: validated business metrics are produced from the Golden layer, not directly from uncleansed operational records.